Complex Social-Ecological Systems: A Practical Introduction
Spring 2026

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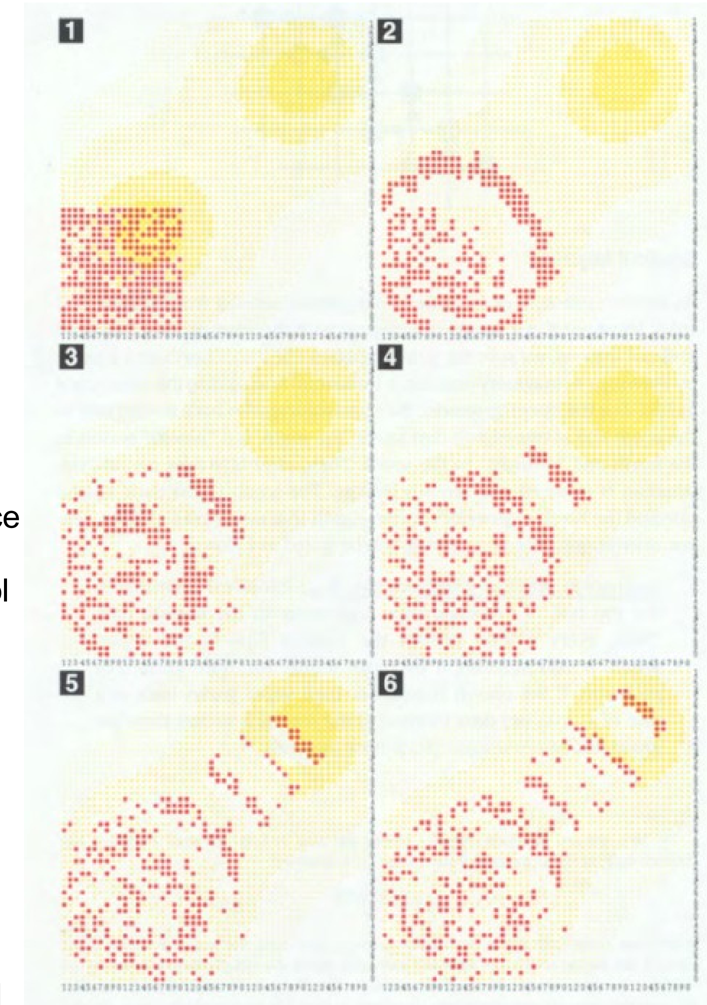
Course Type: **In-Person Lecture**
Office Hours: **TBD**

Course Description



Agents harvest a resource in the Sugarscape model. From Epstein and Axtell, 1996.

A thorough grasp of how social and ecological systems interact is vital if one wants to understand the ways in which human activities shape and are shaped by the environment. This is an especially crucial topic in the age of the Anthropocene and in the face of global climatic change. One of the best tools at our disposal to analyze such systems is the science of *complex* systems, systems "in which large networks of components with no central control and simple rules of operation give rise to complex collective behavior, sophisticated information processing, and adaptation via learning or evolution" (Mitchell, 2009).

Throughout this course, we will return to this seemingly broad and lofty definition with concrete examples and to clarify the way in which human and environmental systems interact. After getting a basic, nontechnical grasp of what exactly a complex system is and why they are relevant to the study of the environment, we will delve into two main methods that are used in the study of such systems: network analysis and agent-based modeling. Students will at this point interact with real-world data and models via graphical interfaces and computer code to see how these tools can be helpful in understanding the way in which human activities lead to environmental changes and vice versa, at both the micro- and macro- scales.

Course Aims

1) To obtain an introductory and non-technical introduction to the study of complex systems, specifically as they relate to social-ecological problems.
2) To broaden students' toolkits for modeling environmental systems by introducing them to network analysis and agent-based modeling.
3) To understand the various assumptions that go into constructing a social-ecological model and what these assumptions mean for model results.

4) To improve students' technical skills with regards to model building and analyzing different types of environmental and social data.

Course Format

The course will consist of in-person lectures along with demonstrations of seminal models and data analytic techniques. Students will be able, throughout the course, to take time to use these models and approaches in class.

Course Prerequisites

There are no formal prerequisites for this course. However, it is expected that students will have a basic grasp of mathematical and statistical concepts and some experience working with data in a computer program, preferably R or Python. That said, the exact level of computational rigor and the programs involved in classwork will be determined by and tailored to enrolled student knowledge.

Course Requirements

<u>Written Homework Assignments (20%)</u>

Three to four (depending on course progress) brief written (mostly nontechnical) homework assignments spaced throughout the course to make sure students are progressing in their knowledge of course topics. Questions will mainly concern course readings and lecture material.

<u>Literature Review (20%)</u>

Students will conduct a review of several papers outside the assigned reading list (or one book-length work). The review will not only summarize the work(s), but make clear connections to problems and methods discussed and demonstrated during the course and raise further research or methodological questions based on the material presented.

<u>Course Project (30%)</u>

Students will work individually or in groups of up to 3 to complete a project based on at least one of the models or methods reviewed in class. This could include, for example, running various simulations of an agent-based model and summarizing key results and insights gleaned from the model. It could also include analyzing a network dataset and interpreting results through the lens of social-ecological problems.

<u>Final Exam (30%)</u>

A brief, cumulative final exam will test students' knowledge of the full breadth of course material. It will mostly be non-technical (very little math or programming knowledge required), and will instead gauge the ability of the students to interpret course concepts and apply them to real-world social-ecological systems.

Required Texts

Complexity: A Guided Tour, Melanie Mitchell, 2009, Oxford University Press.
Agent-Based Modelling & Geographical Information Systems: A Practical Primer, Andrew Crooks et al., 2019, SAGE.

Students are welcome to access these texts in whatever way is easiest and most cost-effective for them. All other readings will be provided on Canvas. Students with difficulty accessing any materials should contact the instructor at the beginning of the course.

Course Plan

<u>Week 1: What is a complex system?</u>
Required readings:
- Mitchell, chs. 1 and 7
- Campbell, *The Great Transition,* pp. 1-19
- Preiser et al., "Social-ecological systems as complex adaptive systems: organizing principles for advancing research methods and approaches"

Topics:
- Definitions of complex and complex adaptive systems
- Quantifying complexity
- Viewing social-ecological systems as complex systems

<u>Week 2: Examples of complex social-ecological systems</u>
Required readings:
- Mitchell, ch. 12
- Carpenter, Brock, and Hanson, "Ecological and social dynamics in simple models of ecosystem management"
- Liu et al., "Systems integration for global sustainability"
- Santa Fe Institute, "Satellite Nilometers and the Global Commons" (video)

Topics
- Complex systems as part of "life"
- Ecosystem dynamics
- Climate change and sustainability
- Agriculture and development

<u>Week 3: Networks and complexity</u>
Required readings:
- Mitchell, ch. 15
- Crooks et al., ch. 8.1-8.2
- Sayles et al., "Social-ecological network analysis for sustainability sciences: a systematic review and innovative research agenda for the future"
- Kreft et al., "Social network data of Swiss farmers related to agricultural climate change mitigation"

Topics:
- What is a network?
- Basic network measures and visualization
- Application of networks to the environmental and social sciences

<u>Week 4: Working with network data</u>
Required readings:

- Mitchell, chs. 16 and 17
- Crooks et al., ch. 8.3-8.6
- Ognyanova, “Network Analysis and Visualization with R and igraph” (website)

Topics
- Network statistics and visualization
- Different network models
- Importing, cleaning, visualizing, and summarizing network data with igraph

<u>Week 5: An introduction to agent-based modeling</u>
Required readings:
- Mitchell, ch. 10
- Crooks et al., ch. 2
- Epstein and Axtell, *Growing Artificial Societies*, ch. 1 (Introduction)

Topics
- What is an ABM?
- What can we use ABMs for?
- Conway’s Game of Life

<u>Week 6: Simple ABMs for social-ecological systems</u>
Required readings:
- Crooks et al., ch. 3
- Epstein and Axtell, *Growing Artifical Societies*, ch. 2 (Life and Death on the Sugarscape)
- Introduction to ABMs in Python (https://mesa.readthedocs.io/stable/getting_started.html) [TBD based on student experience]

Topics
- ABM design principles
- The Sugarscape model
- Basics of building an ABM in Python [TBD based on student experience]

<u>Week 7: Lab week</u>
No new material is presented this week. Instead, students will engage more directly with ABM and network code and, with the instructor’s help, learn to manipulate such code to run simulations and visualize data.

<u>Week 8: More advanced ABMs for social-ecological systems</u>
Required readings:
- Crooks et al., ch. 7
- Epstein and Axtell, *Growing Artifiical Societies*, ch. 2 (Sugar and Spice)
- Wens et al., “Simulating small-scale agricultural adaptation decisions in response to drought risk: An empirical agent-based model for semi-arid Kenya”
- Matthews et al., “Agent-based land-use models: a review of applications”

Topics:
- Refinements to the Sugarscape model
- Adding more complex human behavior to ABMs
- Using advanced ABMs to model complex social-ecological systems
- Integrating ABMs and networks

Week 9: Limitations and the future of ABMs

Required readings:

- Crooks et al., chs. 10 and 12
- Matthias Klusch, “Toward Quantum Computational Agents” (just the introduction and conclusion!)
- David Bentley Hart, “Emergence and Formation”

Topics

- Model validation, with examples in R or Python
- Future outlook for the field
- The philosophy of complex systems and ABMs

Week 10: Conclusion

Required readings:

- Mitchell, ch. 19
- Giorgio Parisi, *In a Flight of Starlings*, ch. 1
- Strogatz, *Nonlinear Dynamics and Chaos*, ch. 1 (Overview) (you can ignore the math)
- Charlotte Higgins, “’There is no such thing as past or future’: physicist Carlo Rovelli on changing how we think about time” (also read the extract from Rovelli’s book)

Topics

- The “edge of chaos” and nonlinear systems
- Entropy and disorder
- Why we study complex systems

Integrity

Please see the Duke Community Standard for guidance on integrity and academic dishonesty issues. Any violation may result in a failing grade and, depending on the severity of the case, could result in additional consequences. For support on plagiarism issues, please consult the Duke Library: https://library.duke.edu/research/plagiarism or other Duke University Resources.

Student Accommodations

Students who wish to be considered for reasonable accommodations for a documented disability or temporary medical condition must self-identify to the Student Disability Access Office (SDAO) by submitting a request for accommodations at https://access.duke.edu/. Accommodations are not applied retroactively so it is recommended that students submit requests to SDAO early.

Harassment and Discrimination

Incidents of harassment and/or discrimination are prohibited at Duke University. All community members have the right to report such incidents without fear of retaliation and seek help and support through campus resources. Sometimes, these incidents may not remain confidential and must be reported to the University. Please consult the University Women’s Center, Student Health, CAPS, University Ombudsperson, and clergy in the Duke Chapel Office. More information about Duke’s policies, processes, and support resources can be found here: https://oie.duke.edu/areas/harassment/ and https://oie.duke.edu/areas/discrimination/

Lauren's Promise: I will listen and believe you if someone threatens you.

AI Policy

AI and complex systems are intimately connected. I thus expect students to be able to understand both the usefulness and limitations of tools like generative AI (e.g. ChatGPT). To this end, although you may use such tools to help you understand concepts and even write code, you may not use it to directly contribute to any final work that you turn in.

Teaching Philosophy and Self-Care

I want this class to be an open space for experimentation and discussion. In particular, I believe that the classroom experience should be not only instructive in a more traditional academic sense, but also hands-on when possible. Hence, I present many opportunities to actually work with data and models in this course. It is my expectation that students will make the most of such opportunities, be curious about what they do not understand, and ask questions.

At the same time, some of the content may be novel and challenging. As is often the case in the environmental sciences, we may be discussing topics that are distressing and/or problematic. Students should prioritize self-care and care for others above all else. Therefore, students should seek health-giving practices as needed during the semester. Counseling is available through Duke Counseling and Psychological Services (CAPS), and additional resources are available through the Duke Women's Center and the Duke Center for Sexual and Gender Diversity. The instructor will, to the extent possible and permitted by the University, also be available to discuss any issues a student may be encountering during the semester that could impact their classwork.